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ST4005CEM Database System

Giriraj Rawat

1) Find students aged between 18 and 21

```sql
select * from developers where age between 18 and 21; -- qn1
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number
1	Aarav Sharma	18	Male	Kathmandu	9800000001
2	Sita Karki	19	Female	Pokhara	9800000002
3	Rohan Thapa	20	Male	Lalitpur	9800000003
4	Anjali Gurung	18	Female	Bhaktapur	9800000004
5	Bikash Rai	21	Male	Dharan	9800000005
6	Pooja Shrestha	19	Female	Biratnagar	9800000006
8	Nisha Magar	20	Female	Butwal	9800000008
10	Ritika Adhikari	18	Female	Hetauda	9800000010
NULL	NULL	NULL	NULL	NULL	NULL

2) Find students whose age is not between 20 and 22

```sql
select * from developers where age  not between 20 and 22; -- qn2
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wr

student_id	student_name	age	gender	city	phone_number
1	Aarav Sharma	18	Male	Kathmandu	9800000001
2	Sita Karki	19	Female	Pokhara	9800000002
4	Anjali Gurung	18	Female	Bhaktapur	9800000004
6	Pooja Shrestha	19	Female	Biratnagar	9800000006
9	Kiran Bhandari	23	Male	Nepalgunj	9800000009
10	Ritika Adhikari	18	Female	Hetauda	9800000010
NULL	NULL	NULL	NULL	NULL	NULL

3) Find students with student_id between 2 and 4

```
select * from developers where student_id between 2 and 4 ;
show create table developers;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number
2	Sita Karki	19	Female	Pokhara	9800000002
3	Rohan Thapa	20	Male	Lalitpur	9800000003
4	Anjali Gurung	18	Female	Bhaktapur	9800000004
NULL	NULL	NULL	NULL	NULL	NULL

4) Find students whose phone number starts between '980' and '985'

```
select * from developers where
phone_number like "980%" or
phone_number like "981%" or
phone_number like "982%" or
phone_number like "983%" or
phone_number like "984%" or
phone_number like "985%";
show create table developers;
```

Result Grid | Filter Rows: | Edit: | Export/In

student_id	student_name	age	gender	city	phone_number
1	Aarav Sharma	18	Male	Kathmandu	9800000001
2	Sita Karki	19	Female	Pokhara	9800000002
3	Rohan Thapa	20	Male	Lalitpur	9800000003
4	Anjali Gurung	18	Female	Bhaktapur	9800000004
5	Bikash Rai	21	Male	Dharan	9800000005
6	Pooja Shrestha	19	Female	Biratnagar	9800000006
7	Suman Tamang	22	Male	Chitwan	9800000007
8	Nisha Magar	20	Female	Butwal	9800000008
9	Kiran Bhandari	23	Male	Nepalgunj	9800000009
10	Ritika Adhikari	18	Female	Hetauda	9800000010

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5) Find students from 'ktm', 'dharan', or 'butwal'

```sql
select * from developers where city in ("ktm","dharan","butwal");
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number
1	ram kc	19	Male	ktm	9801234567
2	hari dc	21	Male	dharan	9812345678
3	sita lama	20	Female	butwal	9823456789
6	nischal shrestha	22	Male	ktm	9856789012
8	suman thapa	21	Male	dharan	9809123456
9	anita gurung	19	Female	butwal	9819988776
10	roshan karki	23	Male	ktm	12345
12	dipa sharma	22	Female	dharan	9851111111
NULL	NULL	NULL	NULL	NULL	NULL

6) Find students not from 'dang' or 'chitwan'

```sql
select * from developers where city not in ("dang","chitwan");
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number
1	ram kc	19	Male	ktm	9801234567
2	hari dc	21	Male	dharan	9812345678
3	sita lama	20	Female	butwal	9823456789
6	nischal shrestha	22	Male	ktm	9856789012
7	bikash adhikari	24	Male	pokhara	9867890123
8	suman thapa	21	Male	dharan	9809123456
9	anita gurung	19	Female	butwal	9819988776
10	roshan karki	23	Male	ktm	12345
12	dipa sharma	22	Female	dharan	9851111111
NULL	NULL	NULL	NULL	NULL	NULL

7) Find students with age 19, 21, or 23

```sql
select * from developers where age in (19,21,23);
```

student_id	student_name	age	gender	city	phone_number
1	ram kc	19	Male	ktm	9801234567
2	hari dc	21	Male	dharan	9812345678
4	rita sah	23	Female	dang	9834567890
8	suman thapa	21	Male	dharan	9809123456
9	anita gurung	19	Female	butwal	9819988776
10	roshan karki	23	Male	ktm	12345
NULL	NULL	NULL	NULL	NULL	NULL

8) Find students whose names are not 'ram kc' or 'hari dc'

```sql
select * from developers where  student_name not like "%ram kc%" and student_name not like "hari dc";
```

student_id	student_name	age	gender	city	phone_number
3	sita lama	20	Female	butwal	9823456789
4	rita sah	23	Female	dang	9834567890
5	gita rai	18	Female	chitwan	9845678901
6	nischal shrestha	22	Male	ktm	9856789012
7	bikash adhikari	24	Male	pokhara	9867890123
8	suman thapa	21	Male	dharan	9809123456
9	anita gurung	19	Female	butwal	9819988776
10	roshan karki	23	Male	ktm	12345
11	lila kc	20	Female	dang	9840000000
12	dipa sharma	22	Female	dharan	9851111111
NULL	NULL	NULL	NULL	NULL	NULL

9) Find students with gender 'm' or 'f' (using IN)

i) Male

```sql
select * from developers where gender in ("male");
```

student_id	student_name	age	gender	city	phone_number
1	ram kc	19	Male	ktm	9801234567
2	hari dc	21	Male	dharan	9812345678
6	nischal shrestha	22	Male	ktm	9856789012
7	bikash adhikari	24	Male	pokhara	9867890123
8	suman thapa	21	Male	dharan	9809123456
10	roshan karki	23	Male	ktm	12345
NULL	NULL	NULL	NULL	NULL	NULL

ii) Female

```sql
select * from developers where gender in ("female");
```

student_id	student_name	age	gender	city	phone_number
3	sita lama	20	Female	butwal	9823456789
4	rita sah	23	Female	dang	9834567890
5	gita rai	18	Female	chitwan	9845678901
9	anita gurung	19	Female	butwal	9819988776
11	lila kc	20	Female	dang	9840000000
12	dipa sharma	22	Female	dharan	9851111111
NULL	NULL	NULL	NULL	NULL	NULL

10) Categorize students as 'Teen' (age <20), 'Young Adult' (20-22), or 'Adult' (23+)

```sql
alter table developers add column status enum("adult","young","teen") after phone_number ;
select * from developers;
update developers set status=case
when age<20 then "teen"
when age between 20 and 22 then "young"
when age >22 then "adult"
end ;
set sql_safe_updates=0;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number	status
1	ram kc	19	Male	ktm	9801234567	teen
2	hari dc	21	Male	dharan	9812345678	young
3	sita lama	20	Female	butwal	9823456789	young
4	rita sah	23	Female	dang	9834567890	adult
5	gita rai	18	Female	chitwan	9845678901	teen
6	nischal shrestha	22	Male	ktm	9856789012	young
7	bikash adhikari	24	Male	pokhara	9867890123	adult
8	suman thapa	21	Male	dharan	9809123456	young
9	anita gurung	19	Female	butwal	9819988776	teen
10	roshan karki	23	Male	ktm	12345	adult
11	lila kc	20	Female	dang	9840000000	young
12	dipa sharma	22	Female	dharan	9851111111	young
NULL	NULL	NULL	NULL	NULL	NULL	NULL

11) Label students from 'ktm' as 'Capital' and others as 'Other City'.

```sql
select student_id,student_name, age, gender,city,phone_number,status ,case
 when city="ktm" then "capital"
else "Other city"
end as label
from developers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	student_name	age	gender	city	phone_number	status	label
1	ram kc	19	Male	ktm	9801234567	teen	capital
2	hari dc	21	Male	dharan	9812345678	young	Other city
3	sita lama	20	Female	butwal	9823456789	young	Other city
4	rita sah	23	Female	dang	9834567890	adult	Other city
5	gita rai	18	Female	chitwan	9845678901	teen	Other city
6	nischal shrestha	22	Male	ktm	9856789012	young	capital
7	bikash adhikari	24	Male	pokhara	9867890123	adult	Other city
8	suman thapa	21	Male	dharan	9809123456	young	Other city
9	anita gurung	19	Female	butwal	9819988776	teen	Other city
10	roshan karki	23	Male	ktm	12345	adult	capital
11	lila kc	20	Female	dang	9840000000	young	Other city
12	dipa sharma	22	Female	dharan	9851111111	young	Other city

12) Create a column showing 'Valid' if phone has 10 digits, else 'Invalid'

```sql

alter table developers add column isValid enum("valid","invalid") ;
update developers set isValid=case
when length(phone_number)=10 then "valid"
else "invalid"
end;
select * from developers;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number	status	isValid
1	ram kc	19	Male	ktm	9801234567	teen	valid
2	hari dc	21	Male	dharan	9812345678	young	valid
3	sita lama	20	Female	butwal	9823456789	young	valid
4	rita sah	23	Female	dang	9834567890	adult	valid
5	gita rai	18	Female	chitwan	9845678901	teen	valid
6	nischal shrestha	22	Male	ktm	9856789012	young	valid
7	bikash adhikari	24	Male	pokhara	9867890123	adult	valid
8	suman thapa	21	Male	dharan	9809123456	young	valid
9	anita gurung	19	Female	butwal	9819988776	teen	valid
10	roshan karki	23	Male	ktm	12345	adult	invalid
11	lila kc	20	Female	dang	9840000000	young	valid
12	dipa sharma	22	Female	dharan	9851111111	young	valid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

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13) Classify students as 'Male', 'Female', or 'Other' based on gender

```sql
select * from developers where gender="male";
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number	status	isValid
1	ram kc	19	Male	ktm	9801234567	teen	valid
2	hari dc	21	Male	dharan	9812345678	young	valid
6	nischal shrestha	22	Male	ktm	9856789012	young	valid
7	bikash adhikari	24	Male	pokhara	9867890123	adult	valid
8	suman thapa	21	Male	dharan	9809123456	young	valid
10	roshan karki	23	Male	ktm	12345	adult	invalid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

14) Mark students as 'Junior' (age <21) or 'Senior' (age ≥21)

```sql
select * from developers where gender="male";
select * , case
when age<21 then "Junior"
else "Senior"
end as experience from developers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	student_name	age	gender	city	phone_number	status	isValid	experience
1	ram kc	19	Male	ktm	9801234567	teen	valid	Junior
2	hari dc	21	Male	dharan	9812345678	young	valid	Senior
3	sita lama	20	Female	butwal	9823456789	young	valid	Junior
4	rita sah	23	Female	dang	9834567890	adult	valid	Senior
5	gita rai	18	Female	chitwan	9845678901	teen	valid	Junior
6	nischal shrestha	22	Male	ktm	9856789012	young	valid	Senior
7	bikash adhikari	24	Male	pokhara	9867890123	adult	valid	Senior
8	suman thapa	21	Male	dharan	9809123456	young	valid	Senior
9	anita gurung	19	Female	butwal	9819988776	teen	valid	Junior
10	roshan karki	23	Male	ktm	12345	adult	invalid	Senior
11	lila kc	20	Female	dang	9840000000	young	valid	Junior
12	dipa sharma	22	Female	dharan	9851111111	young	valid	Senior

15) Find female students from 'ktm' or 'butwal'

```sql
select * from developers where city in ("ktm","butwal") and gender="female";
```

student_id	student_name	age	gender	city	phone_number	status	isValid
3	sita lama	20	Female	butwal	9823456789	young	valid
9	anita gurung	19	Female	butwal	9819988776	teen	valid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

16) Find male students aged 20 or from 'dharan'

```sql
select * from developers where age=20 and city="dharan" and gender="male";
```

student_id	student_name	age	gender	city	phone_number	status	isValid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

17) Find students aged 19-21 who are not from 'dang'

```sql
select * from developers where age between 19 and 21 and city<>"dang";
```

student_id	student_name	age	gender	city	phone_number	status	isValid
1	ram kc	19	Male	ktm	9801234567	teen	valid
2	hari dc	21	Male	dharan	9812345678	young	valid
3	sita lama	20	Female	butwal	9823456789	young	valid
8	suman thapa	21	Male	dharan	9809123456	young	valid
9	anita gurung	19	Female	butwal	9819988776	teen	valid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

18) Find students whose name starts with 'r' and age >20.

```
101 •  select * from developers where age  >=20 and student_name like "r%";
```

Result Grid | Filter Rows: | Edit: | Export/Import: | W

student_id	student_name	age	gender	city	phone_number	status	isValid
4	rita sah	23	Female	dang	9834567890	adult	valid
10	roshan karki	23	Male	ktm	12345	adult	invalid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

19) Find students from 'ktm' or with phone starting with '9809'

```
103 •  select * from developers where city="ktm" or phone_number like "9809%";
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Ce

student_id	student_name	age	gender	city	phone_number	status	isValid
1	ram kc	19	Male	ktm	9801234567	teen	valid
6	nischal shrestha	22	Male	ktm	9856789012	young	valid
8	suman thapa	21	Male	dharan	9809123456	young	valid
10	roshan karki	23	Male	ktm	12345	adult	invalid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

20) Find students with 'sh' in their name

```
105 •  select * from developers where student_name like "%sh%";
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	student_name	age	gender	city	phone_number	status	isValid
6	nischal shrestha	22	Male	ktm	9856789012	young	valid
7	bikash adhikari	24	Male	pokhara	9867890123	adult	valid
10	roshan karki	23	Male	ktm	12345	adult	invalid
12	dipa sharma	22	Female	dharan	9851111111	young	valid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

21) Find students whose name starts with 'r'

```
106 •   select * from developers where student_name like "r%";
```

student_id	student_name	age	gender	city	phone_number	status	isValid
1	ram kc	19	Male	ktm	9801234567	teen	valid
4	rita sah	23	Female	dang	9834567890	adult	valid
10	roshan karki	23	Male	ktm	12345	adult	invalid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

22) Find students whose name ends with 'a'

```
107 •   select * from developers where student_name like "%a";
```

student_id	student_name	age	gender	city	phone_number	status	isValid
3	sita lama	20	Female	butwal	9823456789	young	valid
6	nischal shrestha	22	Male	ktm	9856789012	young	valid
8	suman thapa	21	Male	dharan	9809123456	young	valid
12	dipa sharma	22	Female	dharan	9851111111	young	valid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

23) Find students with 'a' as the second letter in their name

```
108 •   select * from developers where student_name like "_a%";
```

student_id	student_name	age	gender	city	phone_number	status	isValid
1	ram kc	19	Male	ktm	9801234567	teen	valid
2	hari dc	21	Male	dharan	9812345678	young	valid
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

24) Find students whose name doesn't contain 'a'

```
109 •  select * from developers where student_name not like "%a%";
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	student_id	student_name	age	gender	city	phone_number	status	isValid
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

25) Find students with exactly 4 letters in first name (like 'ram kc')

```
110 •  select * from developers where length(SUBSTRING_INDEX(student_name," ",1 ))=4;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

	student_id	student_name	age	gender	city	phone_number	status	isValid
▶	2	hari dc	21	Male	dharan	9812345678	young	valid
	3	sita lama	20	Female	butwal	9823456789	young	valid
	4	rita sah	23	Female	dang	9834567890	adult	valid
	5	gita rai	18	Female	chitwan	9845678901	teen	valid
	11	lila kc	20	Female	dang	9840000000	young	valid
	12	dipa sharma	22	Female	dharan	9851111111	young	valid
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

26) Find students with two-part names (containing space)

```sql
select * from developers where student_name like "% %";
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	student_id	student_name	age	gender	city	phone_number	status	isValid
▶	1	ram kc	19	Male	ktm	9801234567	teen	valid
	2	hari dc	21	Male	dharan	9812345678	young	valid
	3	sita lama	20	Female	butwal	9823456789	young	valid
	4	rita sah	23	Female	dang	9834567890	adult	valid
	5	gita rai	18	Female	chitwan	9845678901	teen	valid
	6	nischal shrestha	22	Male	ktm	9856789012	young	valid
	7	bikash adhikari	24	Male	pokhara	9867890123	adult	valid
	8	suman thapa	21	Male	dharan	9809123456	young	valid
	9	anita gurung	19	Female	butwal	9819988776	teen	valid
	10	roshan karki	23	Male	ktm	12345	adult	invalid
	11	lila kc	20	Female	dang	9840000000	young	valid
	12	dipa sharma	22	Female	dharan	9851111111	young	valid
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

27) Find students whose city name starts with 'd' and has 5 letters

```sql
select * from developers where city like "d%" and length(city)=5;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	student_id	student_name	age	gender	city	phone_number	status	isValid
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

28) Find students whose name has 'i' as the third character

```
113 • select * from developers where student_name like "__i%";
114
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	student_id	student_name	age	gender	city	phone_number	status	isValid
▶	9	anita gurung	19	Female	butwal	9819988776	teen	valid
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

29) Find students whose name doesn't start with 'r' or 's'

```
115 • select * from developers where student_name not like "r%" and student_name not like "s%";
116
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	student_id	student_name	age	gender	city	phone_number	status	isValid
▶	2	hari dc	21	Male	dharan	9812345678	young	valid
	5	gita rai	18	Female	chitwan	9845678901	teen	valid
	6	nischal shrestha	22	Male	ktm	9856789012	young	valid
	7	bikash adhikari	24	Male	pokhara	9867890123	adult	valid
	9	anita gurung	19	Female	butwal	9819988776	teen	valid
	11	lila kc	20	Female	dang	9840000000	young	valid
	12	dipa sharma	22	Female	dharan	9851111111	young	valid
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL